

SI 427 Midsemester Exam
Department of Mathematics, IIT Bombay

Saturday

16:00 - 18:00

September 20, 2025

30 marks

1. Make an index.

2. Answers without justification may not be awarded any marks.

Question 1.

1. A pair of six-sided dice is thrown and the probability that the absolute value of the difference of the values shown by the two dice is 1 is to be determined.

- (a) (1 mark) Write down a suitable discrete probability space (Ω, p) for this experiment.

Solution: We take

$$\Omega = \{(i, j) \mid 1 \leq i, j \leq 6\} \quad \text{and} \quad p(\omega) = \frac{1}{36},$$

for all $\omega \in \Omega$.

(1 mark)

Comment: No marks will be awarded in general unless both Ω and p are defined.

- (b) (1 mark) Write down a suitable random variable $X : \Omega \rightarrow \mathbb{R}$ for this experiment so that $P(X^{-1}(1))$ is the desired probability.

Solution: We define $X((i, j)) = |i - j|$.

(1 mark)

- (c) (1 mark) Compute $P(X^{-1}(1))$.

Solution: We need to determine the number of pairs (i, j) such that $1 \leq i, j \leq 6$ and $|i - j| = 1$. These are exactly the pairs of the form $(i, i + 1)$, $1 \leq i \leq 5$ and $(i, i - 1)$, $2 \leq i \leq 6$. There are 10 such pairs. Hence, $P(X^{-1}(1)) = 10/36 = 5/18$.

2. (3 marks) Three identical coins are kept in a bag. Two of them are unbiased coins in the sense that the probability of getting heads is $1/2$. The remaining one is a biased coin with probability of heads is $3/4$. One coin is selected randomly from the bag, tossed and observed to be the heads up. What is the probability that it is one of the unbiased coins?

Solution: We will call the coins 1, 2 and 3, with coin 3 being biased and let E_1 , E_2 and E_3 be the events that the respective coin has been selected. Let H be the event that “heads” has been observed. We need to calculate $P(E_1 \cup E_2 \mid H)$, or equivalently, $1 - P(E_1 \cup E_2 \mid H) = P(E_3 \mid H)$. **(1 mark)**

By Bayes’ Theorem,

$$P(E_3 \mid H) = \frac{P(H \mid E_3)P(E_3)}{P(H)}.$$

If the problem was not set up properly above, but Bayes’ Theorem has been properly stated and used then one mark will be awarded at this stage.

We are given $P(E_i) = 1/3$ for $1 \leq i \leq 3$. Further, we are given that

$$P(H | E_1) = 1/2, \quad P(H | E_2) = 1/2, \quad P(H | E_3) = 3/4.$$

Hence, $P(H | E_3)P(E_3) = 3/4 \times 1/3 = 1/4$.

(1 mark)

The events E_i , $1 \leq i \leq 3$ are mutually exclusive and exhaustive. Hence,

$$P(H) = P(H \cap E_1) + P(H \cap E_2) + P(H \cap E_3)$$

We know that $P(H \cap E_i) = P(H | E_i)P(E_i)$ for $1 \leq i \leq 3$. Hence,

$$P(H) = 1/2 \times 1/3 + 1/2 \times 1/3 + 3/4 \times 1/3 = 7/12.$$

(1 mark)

Hence, $P(E_3 | H) = 1/4 \times 12/7 = 3/7$. It follows that $P(E_1 \cup E_2 | H) = 1 - 3/7 = 4/7$.

Comment: If a student has calculated $P(E_1 \cup E_2 | H)$ directly using Bayes' Theorem, the same marking scheme has been used. Basically, there is one mark for setting up the problem, one mark for finding the numerator in the expression coming from Bayes' Theorem, and one mark for finding its denominator.

Question 2.

1. (2 marks) Let (Ω, p) be a discrete probability space. Show that if A and B are independent events, then so are their respective complements A' and B' .

Solution:

(1) It is enough to show that A and B' are independent, since we can then apply the statement to the pair B' and A to get the desired result. Now $P(A) = P(A \cap B) + P(A \cap B')$, since B and B' are mutually exclusive and exhaustive events. **(1 mark)**

Hence, $P(A) = P(A)P(B) + P(A \cap B')$, since A and B are independent. It follows that

$$P(A \cap B') = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B').$$

(1 mark)

(2) We have

$$P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B) = 1 - [P(A) + P(B) - P(A \cap B)].$$

(1 mark)

Since A and B are independent, this gives

$$1 - P(A) - P(B) + P(A)P(B) = (1 - P(A))(1 - P(B)) = P(A')P(B').$$

(1 mark)

2. (2 marks) In a population of 10 people, 3 are left-handed and 7 are right-handed. Let X be the random variable that counts the number of left-handed people in a randomly drawn sample of 3 people. Let f_X be the associated probability mass function. Calculate $f_X(2)$.

Solution: Clearly $X \sim \text{Hypergeo}(3, 7, 3)$. The number of ways of drawing a random sample of 3 people is $\binom{10}{3}$. The number of ways in which exactly 2 of these are left-handed is $\binom{3}{2}\binom{7}{1}$.

(1 mark)

Thus,

$$f_X(2) = \frac{\binom{3}{2}\binom{7}{1}}{\binom{10}{3}} = \frac{3 \cdot 7 \cdot 3!}{10 \cdot 9 \cdot 8} = \frac{7}{40}.$$

(1 mark)

3. (2 marks) With notation as in the previous question, calculate $E[X]$. Proceeding as before, for any $0 \leq k \leq 3$,

$$f_X(k) = \frac{\binom{3}{k}\binom{7}{7-k}}{\binom{10}{3}}.$$

(1 mark)

We have

$$\begin{aligned} E[X] &= \sum_{k=0}^3 k f_X(k) = \sum_{k=0}^3 k \frac{\binom{3}{k} \binom{7}{7-k}}{\binom{10}{3}} \\ &= \frac{9}{10} \sum_{k=1}^3 \frac{\binom{2}{k-1} \binom{7}{7-k}}{\binom{10}{3}} \\ &= \frac{9}{10} \sum_{j=0}^2 \frac{\binom{2}{j} \binom{6}{6-j}}{\binom{9}{2}} = \frac{9}{10}. \end{aligned}$$

(1 mark)

Recall that the sum in the penultimate term is 1 since it is the sum of the probabilities of choosing 0, 1 or 2 left-handed people in a randomly chosen sample of 2 people.)

(If there is a calculation mistake towards the end, one 1/2 mark has been awarded.)

Question 3.

1. (3 marks) Determine if the function

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \text{ and} \\ 1 - e^{-x^2} & \text{if } x \geq 0 \end{cases}$$

is a CDF. If it is, determine its probability density function (PDF).

Solution: The function e^{-x^2} is a decreasing function, so $1 - e^{-x^2}$ is an increasing function (we know that $(e^{-x^2})' = -2xe^{-x^2}$ which is always negative). **(1 mark)**

Clearly $\lim_{x \rightarrow \infty} 1 - e^{-x^2} = 1$ (even more obviously $\lim_{x \rightarrow \infty} F(x) = 1$, a half mark will be awarded even if this is not explicitly stated). **(1/2 mark)**

The given function is continuous (including at 0), so it is necessarily right-continuous. **(1/2 mark)**

The PDF is

$$f(x) = 2xe^{-x^2}.$$

(1 mark)

2. (a) (1 mark) Write down the probability density function (PDF) for the Gamma distribution with shape parameter $\nu > 0$ and scalar parameter $\lambda > 0$.

Solution: The PDF is

$$f(t) = \begin{cases} 0 & \text{if } t < 0, \text{ and} \\ \frac{\lambda^\nu t^{\nu-1} e^{-\lambda t}}{\Gamma(\nu)} & \text{if } t \geq 0. \end{cases}$$

(1 mark)

- (b) (2 marks) When $\nu = 2$ write down an expression for its CDF.

Solution: Integrate by parts (repeatedly) to get,

$$F(t) = 1 - e^{-\lambda t} \sum_{k=0}^1 \frac{(\lambda t)^k}{k!} = 1 - e^{-\lambda t} - \lambda t e^{-\lambda t}$$

(2 marks)

It is difficult to give partial credit here, but if someone has realised that they should use integration by parts but then made a silly mistake, they have been awarded 1 mark.

Question 4.

1. (2 marks) Define an algebra $\mathcal{F}$ on a set Ω .

Solution: There are four axioms that usually define an algebra, but only three are necessary (closure under either unions or intersections is enough). If three axioms are correct 2 marks have been awarded. Otherwise, one 1/2 mark per axiom.

2. (2 marks) Define a monotone class on a set Ω .

Solution: There is only one axiom for a monotone class. Stating it correctly yields 2 marks.

3. (2 marks) Show that an algebra $\mathcal{F}$ which is a monotone class is a σ -algebra.

Solution: Let $\mathcal{F}$ be an algebra which is also a monotone class. We need to show that if A_n , $n \in \mathbb{N}$ is a collection of pairwise disjoint sets in $\mathcal{F}$, $\cup_{n=1}^{\infty} A_n \in \mathcal{F}$. **(1 mark)**

Let $B_n = \cup_{k=1}^n A_k$ for $n \in \mathbb{N}$. Then $B_n \subseteq B_{n+1}$ for all $n \in \mathbb{N}$, so B_n is a non-decreasing sequence of sets in $\mathcal{F}$, and $\cup_{n=1}^{\infty} A_n = \cup_{n=1}^{\infty} B_n$.

Since $\mathcal{F}$ is a monotone class, $\cup_{n=1}^{\infty} B_n \in \mathcal{F}$. **(1 mark)**

Question 5.

1. (2 marks) Let $\mathcal{B}$ be the Borel σ -algebra on $\mathbb{R}$. Let $\delta : \mathcal{B} \rightarrow [0, 1]$ be defined by

$$\delta(E) := \begin{cases} 1 & \text{if } 0 \in E, \text{ and} \\ 0 & \text{otherwise.} \end{cases}$$

Show that δ is a countably additive probability measure on $\mathcal{B}$.

Solution: Since $0 \in \mathbb{R}$, $\delta(\mathbb{R}) = 1$. (1/2 mark)

Let E_n , $n \in \mathbb{N}$ be a collection of pairwise disjoint subsets. Two cases arise. If $0 \in E_k$ for some $k \in \mathbb{N}$, $\delta(E_k) = 1$ and $\delta(E_n) = 0$ if $n \neq k$. Since $0 \in (\cup_{n=1}^{\infty} E_n)$,

$$\delta(\cup_{n=1}^{\infty} E_n) = 1 = \sum_{n=1}^{\infty} \delta(E_n).$$

If $0 \notin E_k$ for any $k \in \mathbb{N}$, $0 \notin (\cup_{n=1}^{\infty} E_n)$, so

$$\delta(\cup_{n=1}^{\infty} E_n) = 0 = \sum_{n=1}^{\infty} \delta(E_n),$$

since $\delta(E_n) = 0$ for all $n \in \mathbb{N}$.

2. (2 marks) What is the cumulative distribution function (CDF) F of δ ?

Solution: We know that $P([a, b])$ should be given by $F(b) - F(a)$. If $0 \in [a, b]$, $F(b) - F(a) = 1$. Since $0 \leq F(x) \leq 1$ for all $x \in \mathbb{R}$, we must have $F(b) = 1$ and $F(a) = 0$. Thus, for every $a < 0$, $F(a) = 0$ and for every $b > 0$, $F(b) = 1$. Since F is required to be right continuous, $F(0) = 1$. Thus, if F is the function defined by

$$F(t) = \begin{cases} 0 & \text{if } t < 0, \text{ and} \\ 1 & \text{if } t \geq 0 \end{cases}$$

we see that it satisfies the necessary conditions. Clearly, δ arises from this CDF.

3. (2 marks) Can the function F in the previous exercise be the CDF of a discrete random variable?

Solution: This can be viewed as the special case of the Bernoulli distribution when $p = 0$. We can take (Ω, p) to be any discrete probability space with $p(\omega_0) = 1$ for some $\omega_0 \in \Omega$ and $p(\omega) = 0$ if $\omega \neq \omega_0$. We define $X : \Omega \rightarrow \mathbb{R}$ to be the function

$$X(\omega) = \begin{cases} 0 & \text{if } \omega \neq \omega_0 \\ 1 & \text{if } \omega = \omega_0. \end{cases}$$

Then $F_X(t) = F(t)$